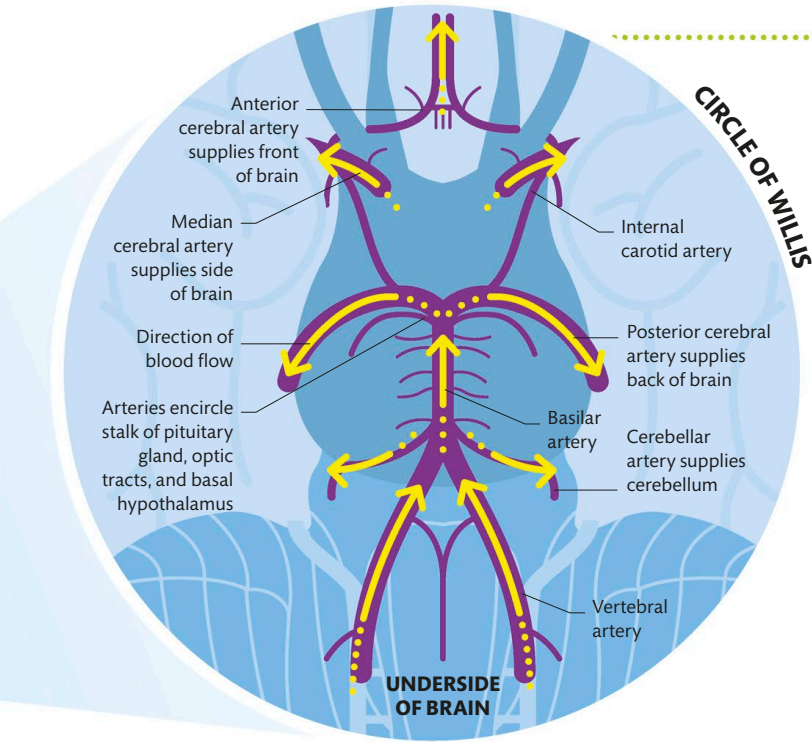




GLUCOSE FUEL

The human brain makes up just 2 percent of the body's total weight, but it consumes 20 percent of its energy. The large human brain is an expensive organ to run, but the benefits of a big, smart brain make it a good investment.



**BRAIN
SIZE: 2%**



**BRAIN'S ENERGY
NEEDS: 20%**

The Circle of Willis

The carotid and vertebral supplies connect at the base of the brain, via communicating arteries, to create a vascular loop called the Circle of Willis. This feature ensures cerebral blood flow is maintained, even if one of the arteries is blocked.

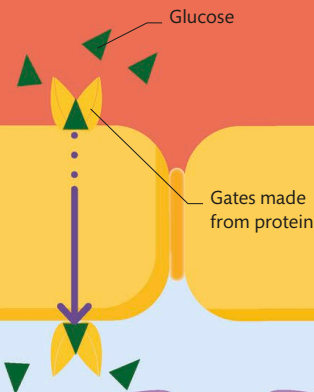


**THE BODY'S ENTIRE
SUPPLY OF BLOOD IS
PUMPED THROUGH THE
BRAIN EVERY 7 MINUTES**



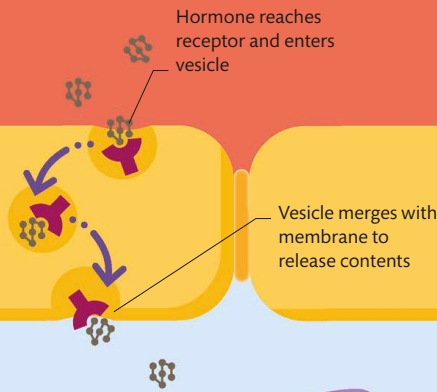
Protein transporters

Glucose and other essential molecules are actively moved across the barrier through channels and gates in the membrane.



Receptors

Hormones and similar substances are picked up by receptors. They are enclosed in a vesicle (sac) of membrane for passage through the cell.



Transcytosis

Large proteins, which are too big to pass through channels, are absorbed by the membrane and enclosed in a vesicle for its journey through the cell.



Active efflux

When unwanted materials diffuse through the blood-brain barrier, they are removed by a biochemical pumping system called efflux transporters.

